

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B. Tech (CBCS) Degree Examinations
 Summer Semester July 2024
CV2103-1 – STRUCTURAL ANALYSIS-I

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all Twenty questions in the OMR Sheet provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- If in a pin-jointed plane frame $(m + r) > 2j$, then the frame is
 - Stable and statically determinate
 - Stable and statically indeterminate
 - Unstable
 - None of the above
- Which of the following methods of structural analysis is a force method?
 - Slope deflection method
 - Column analogy method
 - Moment distribution method
 - None of the above
- The fixed support in a real beam becomes in the conjugate beam a
 - Roller support
 - Hinged support
 - Fixed support
 - Free end
- A rigid-jointed plane frame is stable and statically determinate if
 - $(m + r) = 2j$
 - $(m + r) = 3j$
 - $(3m + r) = 3j$
 - $(m + 3r) = 3j$
- Which of the following is correct with reference to conjugate beam
 - $\Delta = \frac{\partial U}{\partial P}$
 - $\Delta = \frac{\partial P}{\partial U}$
 - $\Delta = \frac{\partial U}{\partial M}$
 - $\Delta = \frac{\partial M}{\partial P}$
- The degree of static indeterminacy of a rigid-jointed space frame is
 - $m + r - 2j$
 - $m + r - 3j$
 - $3m + r - 3j$
 - $6m + r - 6j$
- Degree of kinematic indeterminacy of a pin-jointed plane frame is given by
 - $2j - r$
 - $j - 2r$
 - $3j - r$
 - $2j + r$
- While using three moments equation, a fixed end of a continuous beam is replaced by an additional span of
 - Zero length
 - Infinite length
 - Zero moment of inertia
 - None of the above
- Castigliano's first theorem is applicable
 - For statically determinate structures only
 - When the system behaves elastically
 - Only when principle of superposition is valid
 - None of the above
- The Castigliano's second theorem can be used to compute deflections
 - In statically determinate structures only
 - For any type of structure
 - At the point under the load only
 - For beams and frames only
- The three moments equation is applicable only when
 - The beam is prismatic
 - There is no settlement of supports
 - There is no discontinuity such as hinges within the span
 - The spans are equal

12. In unit load method, the load should be applied in the direction of
 A) deflection B) Perpendicular to deflection
 C) vertical D) Horizontal
13. The number of independent equations to be satisfied for static equilibrium of a plane structure is
 A) 1 B) 2
 C) 3 D) 6
14. The number of independent displacement components at each joint of a rigid-jointed space frame is
 A) 1 B) 2
 C) 3 D) 6
15. For a two-hinged arch, if one of the supports settles down vertically, then the horizontal thrust
 A) Is increased B) Is decreased
 C) Remains unchanged D) Becomes zero
16. For a symmetrical two hinged parabolic arch, if one of the supports settles horizontally, then the horizontal thrust
 A) Is increased B) Is decreased
 C) Remains unchanged D) Becomes zero
17. What is the bending moment on any section of a cable
 A) 0 B) Depends on span
 C) Depends on dip D) Depends on cable diameter
18. In an arch bending moment at any section is less than that for a beam of same span, loading and end conditions because of
 A) Vertical support reaction B) Horizontal support reaction
 C) Moment reaction at support D) None of these
19. A three hinged arch is
 A) Statically determinate B) Statically indeterminate
 C) Depends on loading D) Depends on span
20. A two hinged arch is
 A) Statically determinate B) Statically indeterminate
 C) Depends on loading D) Depends on span

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Define (i) degree of static indeterminacy (ii) degree of kinematic indeterminacy
 b) Mention the assumptions made in the analysis of truss.
 c) Determine the slope and vertical displacement at the free end "B" of the cantilever beam shown in Figure. Q3(c) using conjugate beam method.

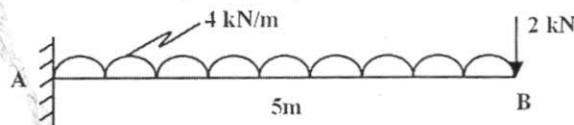


Figure.Q3(c)

Marks	BT*	CO*	PO*
4	L*1	1	1
4	L1	1	1
8	L3	2	2

2. a) Analyze the truss shown in Figure. Q2(a) by method of joints and tabulate the member forces.

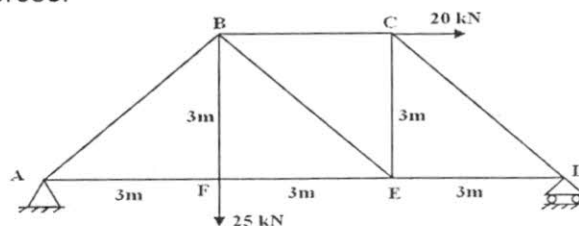


Figure. Q2(a)

- b) Determine the slope at point B and deflection at point C for the simple supported beam shown in Figure. Q2(b) by conjugate beam method.

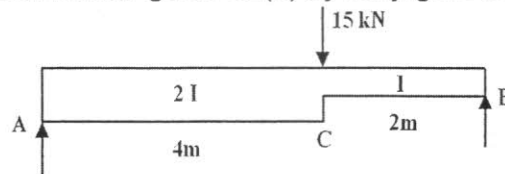


Figure Q2(b)

8 L3 2 2

3. a) Determine the forces in all the members of the truss shown in Figure.Q3(a) by method of joints and tabulate the results.

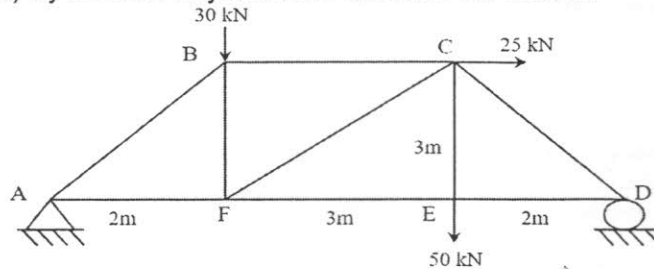


Figure.Q3(a)

8 L2 1 2

- b) Determine the slope and deflection at the free end of the cantilever beam shown in the figure by Conjugate Beam method.

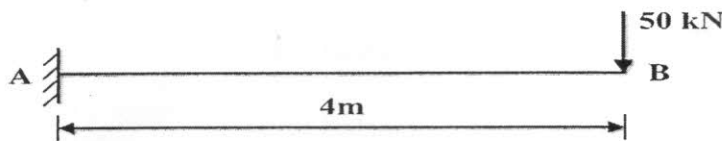


Figure.Q3(b)

8 L3 2 2

Unit – II

4. a) Determine the vertical deflection at the free end of the cantilever beam shown in Figure.Q4(a) using Castigliano's First Theorem.

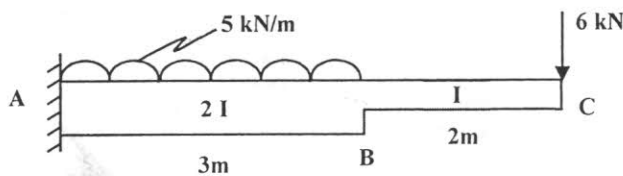


Figure.Q4(a)

8 L2 3 2

- b) Analyze the continuous beam ABC shown in Figure.Q4(b) using three moment equation and draw the bending moment diagram.

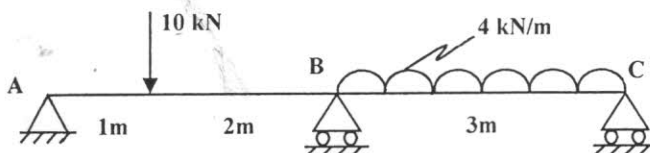


Figure.Q4(b)

8 L3 4 2

5. a) Determine the vertical deflection at point C for the overhanging beam shown in Figure.Q5(a) using Unit Load Method.

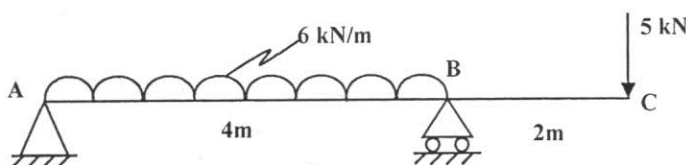


Figure.Q5(a)

8 L2 3 2

- b) Analyze the propped cantilever beam shown in Figure.Q5(b) using three moment equation and draw the bending moment diagram.

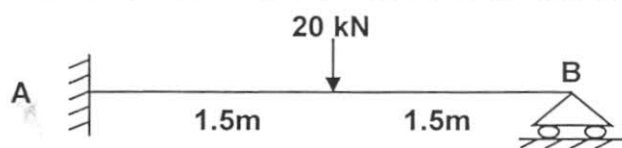


Figure.Q5(b)

8 L3 4 2

- a) Using Castigliano's theorem (dummy load method), determine deflection at mid-span of simple supported beam shown in Figure.Q6(a).

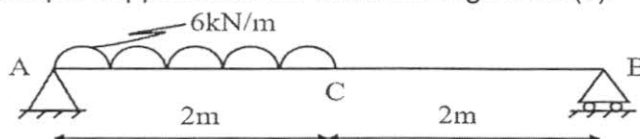


Figure.Q6(a)

8 L2 3 2

- b) Using three moment equation, analyze the continuous beam ABCD shown in Figure.Q6(b) and draw BMD.



Figure.Q6(b)

8 L2 4 2

Unit – III

7. a) A three hinged parabolic arch of span 24m and rise 6m is subjected to a point load of 50kN and 150 kN at a distance of 8m and 20m respectively from the left support. Determine (i) reactions at supports, (ii) resultant reactions and its inclination w.r.t horizontal, (iii) normal thrust and radial shear at a distance of 6m from the left support.
8. a) A cable of uniform section suspended between two supports 80m apart carries a UDL of 12kN/m throughout the span and has a central dip of 8m. Determine the maximum and minimum tension in the cable, length of the cable and minimum cross-sectional area of cable if allowable stress is 280Mpa.

16 L3 5 2

16 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
